ME2201: Kinematics and Dynamics of Machinery

Tutorial 2

Submission Date: 12-Aug-2026, in class

Answer any two of the three questions given below

1. Consider the 3R-1P mechanism shown in Figure 1.
   a. Which link in the mechanism must be fixed such that the inversion corresponds to an eccentric slider crank or an eccentric slider-rocker mechanism?
   b. For the inversion in part a, determine the condition in terms of a, b, c, and d for obtaining an eccentric slider-crank mechanism
   c. Consider the following dimensions for the mechanism: a = 7 cm, b = 5 cm, c = 4 cm, and d = 1 cm. Determine the maximum angles through which links 2 and 4 rotate when link 1 is fixed.

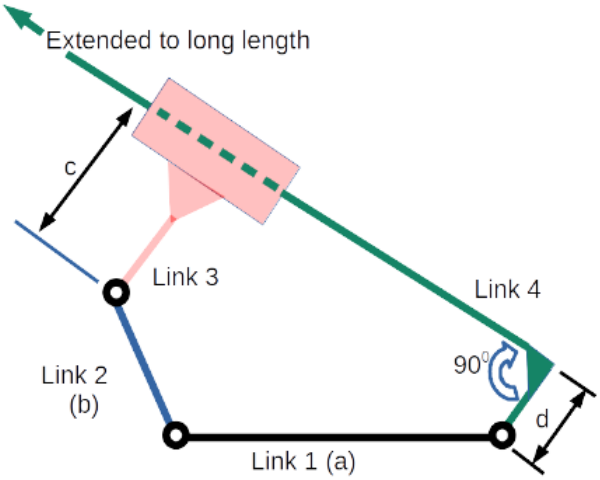


Figure 1: 3R-1P mechanism

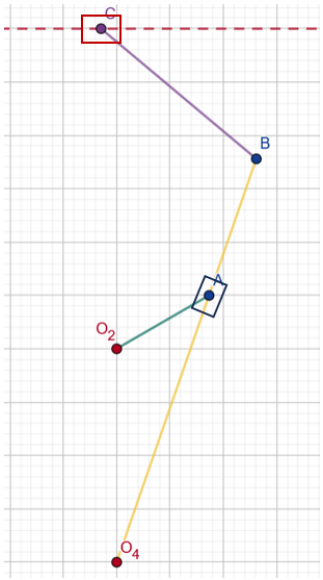


Figure 2: Quick return mechanism

2. Based on the kinematic diagram of the slotted-lever quick return mechanism shown in Figure 2, what are the stroke length and quick return ratio? Link lengths are as follows: $O_2O_4 = 2$, $O_2A = 1$, $O_4B = 4$, BC = 1.9, and the vertical distance between O_2 and the slider path is 2. All dimensions are in cm.
3. Consider the scaled image of a JCB 3DX backhoe loader shown on the next page. Draw the kinematic diagram of the backhoe attachment shown in the figure. If actuator 1 is retracted to half its length and actuator 2 is extended to one and a quarter times its length, determine the change in the bucket's orientation.

12-Aug-2026

